Elastic Deformation of Two-Dimensional Objects

Combined working draft (sections drafted so far)

Group 02

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1 Building the model

We study how the geometry of a two-dimensional elastic object and its supports affect its response to applied forces. Our unknown is the displacement $\mathbf{u}(x, y)$, which tells us how far each point moves horizontally and vertically. Differences in movement between nearby points describe local stretching and shearing, called strain. The material resists these changes through internal forces per unit area, called stress. We assume small deformations and a linear relation between strain and stress, leaving the material choice to the individual cases [4].

We look for the shape after the object has settled under the load. Forces must then balance in both directions throughout the material:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + g_x = 0, \qquad \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + g_y = 0. \tag{1}$$

Here the σ terms are stress components and g_x, g_y are applied forces per unit volume. Since stress depends on strain, and strain depends on displacement, Eq. (1) gives equations for $\mathbf{u}$. We also specify where forces act on the boundary and which movements the supports prevent.

To make this a finite problem, we divide the region into filled triangles as shown in Fig. 1, following its boundary and leaving any holes empty [10]. Their vertices are the mesh nodes, each with two unknown displacement components. Inside each triangle, displacement varies linearly between its three vertex values. Adjacent triangles share these values, so their displacements agree along the common edge. This represents movement throughout the solid; the triangle edges are not physical bars. Smaller triangles allow us to capture more variation in the deformation.

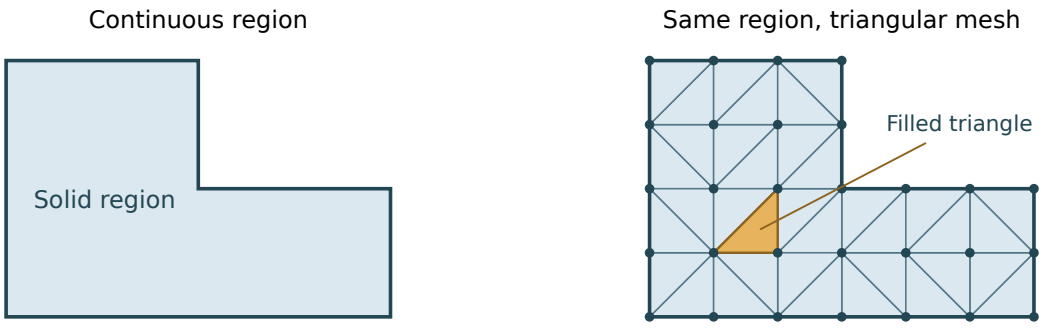


Figure 1: Triangulation of an illustrative region. The filled triangles cover the same solid area and share nodes along their edges. Displacement is approximated throughout each triangle using its three vertex values. The outline is an example, not a selected project geometry.

The finite element method applies Eq. (1) by balancing internal and external work, meaning forces acting through small allowed displacements. Using the triangle approximation and adding

contributions at shared nodes gives

$$Kd = f + r, \tag{2}$$

where d collects node displacements, f the applied forces and r the forces exerted by supports. The stiffness matrix K describes how the mesh and material resist displacement. Most entries are zero because only nodes in a common triangle interact directly.

Finally, we separate unknown displacements (F) from values prescribed at supports (C). The free rows of Eq. (2) become

$$Ax = b. \tag{3}$$

Here $A = K_{FF}$, $x = d_F$ and $b = f_F - K_{FC}d_C$; for fixed supports, $d_C = 0$ and $b = f_F$. Supports must prevent each separate piece from freely sliding or turning. Solving Eq. (3) then gives the displacements used to draw the deformed object.

2 Displacement, forces and matrix construction

We now build the system $Ax = b$ of the preceding section in three steps: the material's response, each triangle's contribution and the supports.

2.1 From displacement to internal forces

We model a thin sheet of constant thickness $t > 0$, made of one material with the same properties everywhere and in every direction. Loads act in its plane and its faces are free, so we use *plane stress*: stresses through the thickness are neglected. Loading is static, and strains and rotations are small enough for a linear model; bending and permanent deformation are excluded.

The material resists stretching and changes of angle. For displacement $\mathbf{u} = (u_x, u_y)^T$, these changes form the strain vector

$$\varepsilon(\mathbf{u}) = \begin{bmatrix} \partial u_x/\partial x \\ \partial u_y/\partial y \\ \partial u_x/\partial y + \partial u_y/\partial x \end{bmatrix}, \qquad \sigma = D\varepsilon, \qquad D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}. \tag{4}$$

The first two entries measure stretching in the coordinate directions; the third measures the change of a right angle (engineering shear). The stress vector $\sigma = (\sigma_{xx}, \sigma_{yy}, \sigma_{xy})^T$ contains the corresponding internal forces per unit area. Young's modulus $E > 0$ controls resistance to stretching, while Poisson's ratio $-1 < \nu < 1/2$ describes the sideways response [1]. Lengths are in metres and stress and E in pascals.

2.2 From triangles to the stiffness matrix

Inside triangle e, we write displacement as $\mathbf{u}_h = \sum_{i=1}^{3} N_i \mathbf{u}_i$. Each linear weight N_i is one at vertex i and zero at the other vertices. Differentiating these weights gives a matrix B_e that converts the triangle's six vertex displacements, collected in d_e, into strain: $\varepsilon(\mathbf{u}_h) = B_e d_e$. The derivatives are constant, so strain is constant within each triangle.

We express force balance by equating internal and external work for any small test movement allowed by the supports. Substituting the triangle approximation gives its stiffness matrix

$$K_e = \int_{T_e} tB_e^T D B_e \, dA = tS_e B_e^T D B_e, \tag{5}$$

where T_e is the triangle and $S_e > 0$ its area. This 6×6 matrix converts vertex movements into internal nodal forces [15]. We add its entries to the corresponding rows and columns of K, using the displacement order $d = (u_{x,0}, u_{y,0}, \ldots, u_{x,n-1}, u_{y,n-1})^T$ for n nodes. Contributions at shared nodes are added together. Nodes without a common triangle have no direct interaction, so most entries are zero and we store K sparsely.

The square of Section 4 uses a structured mesh of squares cut along a diagonal; the other objects are meshed with Triangle [10] (Section 5). Every mesh has positive-area triangles, matching shared edges and empty holes.[1]

External forces are distributed to nodes using the same weights N_i. For example, a constant boundary force per unit area $\bar{\tau}$, called traction, gives a force per unit edge length $q = t\bar{\tau}$. An edge of length ℓ contributes $\ell q/2$ to each endpoint [15]. The load vector f is in newtons and K in N/m. Under traction loading both K and f are proportional to t, so the displacement does not depend on the thickness.

2.3 From supports to a solvable system

The supports prescribe displacement components indexed by C; the remaining components, indexed by F, are unknown. A support can prevent movement in one direction while allowing it in the other. Taking the free rows of $Kd = f + r$ and moving known terms to the right gives

$$\underbrace{K_{FF}}_{A}\underbrace{d_F}_{x} = \underbrace{f_F - K_{FC}d_C}_{b}. \tag{6}$$

The subscripts select rows and columns, respectively. This recovers $Ax = b$ from Section 1; fixed supports have $d_C = 0$ and $b = f_F$. After solving, the forces exerted by supports, called reactions, are

$$r_C = K_{CF}d_F + K_{CC}d_C - f_C, \tag{7}$$

using the original loads, including any applied at supports. Applied forces and reactions together must balance total force and moment.

The construction also explains why supports are essential. Since D is symmetric and positive definite, K is symmetric and

$$d^T K d = \sum_e t S_e (B_e d_e)^T D (B_e d_e) \geq 0. \tag{8}$$

Half this quantity is the stored elastic energy. An unsupported piece can slide or rotate without straining, giving zero energy and making K singular. If the supports prevent both translations and the rotation of every connected piece, the energy is positive for every nonzero free displacement. Thus A is symmetric positive definite.

How strongly the displacement reacts to small load changes is measured by the condition number

$$\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}, \tag{9}$$

the ratio of its largest and smallest eigenvalues. A large ratio means a small percentage change in load can cause a much larger percentage change in displacement. These properties guide the solver choices and accuracy checks in the next section.

3 Numerical methods and comparison protocol

Section 2 showed that A is sparse, symmetric and positive definite, with sensitivity measured by $\kappa_2(A)$. These properties decide which solvers apply and how fast they converge. Let $N = |F|$ be the number of unknowns, and x_k, $e_k = x - x_k$ and $r_k = b - Ax_k$ the k-th iterate, error and residual.

A direct method factorizes A as $PA = LU$ or, for symmetric positive definite A, as $A = LL^T$ (Cholesky) at half the cost, then solves two triangular systems [14]; forming A^{-1} costs more and

[1] We use our own assembly routine (`fem/elasticity.py`); the course-provided routine was not available.

is less accurate. Iterative methods avoid the extra nonzeros (fill-in) of sparse factors and only multiply by A.

A stationary iteration corrects x_k with the residual, using an easily inverted approximation M of A: $x_{k+1} = x_k + M^{-1} r_k$, so $e_{k+1} = (I - M^{-1}A)\, e_k$. It therefore converges for every x_0 if and only if the spectral radius $\rho(I - M^{-1}A)$ is below one [7]. Jacobi uses $M = D_A$, the diagonal of A. Its iteration matrix is similar to $I - S$ with $S = D_A^{-1/2} A D_A^{-1/2}$, so

$$\rho_J = \max_i |1 - \lambda_i(S)|. \tag{10}$$

As $\lambda_i(S) > 0$, Jacobi converges if and only if $\lambda_{\max}(S) < 2$. Damped Jacobi, $M = D_A/\omega$, converges if and only if $0 < \omega < 2/\lambda_{\max}(S)$; we use $\omega = 0.5$. Gauss–Seidel takes M as the lower triangle of A and converges for every symmetric positive definite A [7].

The remaining methods exploit that x minimizes $\phi(y) = \tfrac{1}{2} y^T A y - b^T y$, the elastic energy of Eq. (8) minus the work of the loads. Steepest descent minimizes ϕ along $r_k = -\nabla\phi(x_k)$; each step multiplies the energy norm of the error, $\|e\|_A = (e^T A e)^{1/2}$, by at most $(\kappa - 1)/(\kappa + 1)$, where $\kappa = \kappa_2(A)$. The conjugate gradient (CG) method instead minimizes $\|e_k\|_A$ over $x_0 + \mathcal{K}_k$, where $\mathcal{K}_k = \operatorname{span}\{r_0, A r_0, \ldots, A^{k-1} r_0\}$ is the Krylov space. It needs one product by A per step and satisfies [11]

$$\|e_k\|_A \le 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|e_0\|_A, \tag{11}$$

so its iteration count grows like $\sqrt{\kappa}$ rather than κ. Diagonally preconditioned CG (PCG) is equivalent to CG applied to S, so $\kappa_2(S)$ replaces κ; it helps only if the diagonal of A varies strongly. MINRES minimizes $\|r_k\|_2$ over the same space [5]. Full GMRES does so for any A but stores all k basis vectors [6], so it shows the cost of ignoring symmetry.

Comparison protocol. Every method receives the same A and b, starts from $x_0 = 0$ and is accepted only if its recomputed relative residual satisfies

$$\eta(\hat{x}) = \frac{\|b - A\hat{x}\|_2}{\|b\|_2} < 10^{-8}. \tag{12}$$

Reaching the cap of 20,000 iterations, or $\eta > 10^{10}$ for ordinary Jacobi, counts as failure. Accuracy is the relative error $\|\hat{x} - x_{\mathrm{ref}}\|_2 / \|x_{\mathrm{ref}}\|_2$ against the sparse LU solution x_{ref} of the same system [9]. Times are the best of three runs, excluding assembly and diagnostics. Extreme eigenvalues, for $\kappa_2(A)$ and ρ_J, come from dense or, for large systems, sparse eigensolvers [8].

Since $\hat{x}$ solves exactly the system with load perturbed by its residual, the standard bound [14]

$$\frac{\|\delta x\|_2}{\|x\|_2} \le \kappa_2(A) \frac{\|\delta b\|_2}{\|b\|_2} \tag{13}$$

limits its relative error to $\kappa_2(A)\,\eta$. This *algebraic* error measures how well one mesh's equations are solved. *Discretization* error, between finite element and continuum displacements, does not shrink with the tolerance; we estimate it by comparing meshes at matched physical points.

Later sections also check that d_C equals the prescribed values, that applied loads and the reactions of Eq. (7) have zero resultant force and moment, and that the response to a random load perturbation obeys Eq. (13). A patch test pulls a plate held only by rollers (left edge fixed horizontally, corner at the origin vertically) with uniform traction p. Its exact solution $u_x = px/E$, $u_y = -\nu p y/E$ is linear, so linear triangles must reproduce it to rounding on any mesh, verifying assembly, loading and reduction.

4 Simple square demonstration

We now apply the construction of Section 2 and the protocol of Section 3 to the simplest case we can check by hand: a square plate pulled uniformly at one end.

The plate occupies $0 \le x, y \le L$ with $L = 1$ m, thickness $t = 0.01$ m, $E = 200$ GPa and $\nu = 0.3$, in plane stress (Fig. 2a). The left edge is clamped, $u_x = u_y = 0$, and the right edge carries the uniform traction $\bar{\tau} = (p, 0)$ with $p = 100$ MPa, a total force of $ptL = 1$ MN. The top and bottom edges are free. The mesh has $m \times m$ square cells, each cut along one diagonal, giving $(m+1)^2$ nodes and $2m^2$ triangles. The clamp fixes $2(m+1)$ displacement components, leaving $N = 2m(m+1)$ unknowns. A free bar under this load would stretch by $pL/E = 0.5$ mm, so we expect the right edge to move by about this amount.

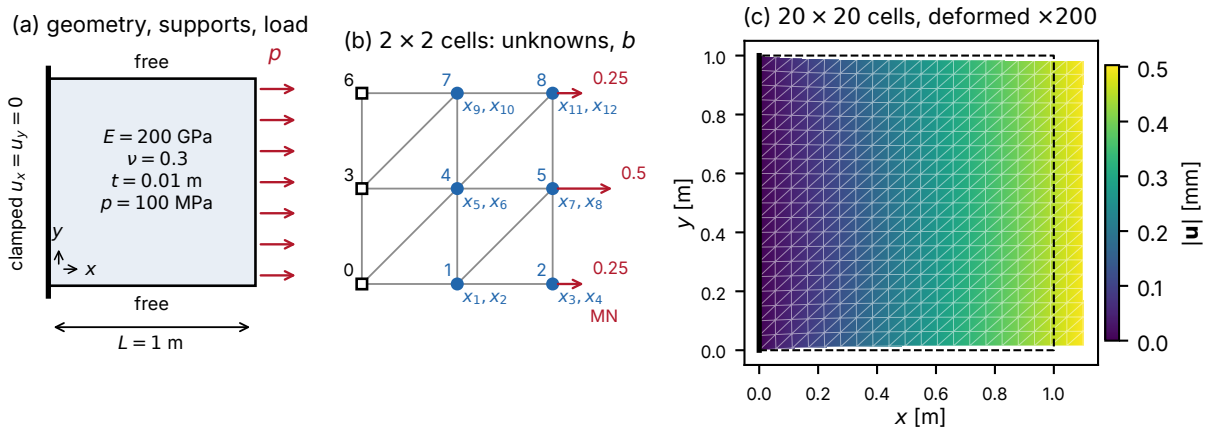


Figure 2: (a) Geometry, supports and load. (b) The 2×2-cell mesh: clamped nodes (squares), free nodes (dots) with their unknowns $x_1, \ldots, x_{12}$, and the nodal forces forming b. (c) Original outline (dashed) and deformed 20×20-cell plate, displacements amplified by 200; colour is the displacement magnitude in mm.

4.1 A small system solved directly

For $m = 2$ there are 9 nodes and 8 triangles. Clamping nodes 0, 3 and 6 leaves $N = 12$ unknowns, two per free node (Fig. 2b). As the prescribed displacements are zero, $b = f_F$. Each half of the right edge carries 0.5 MN, shared equally by its two end nodes, so the right-edge nodes receive 0.25, 0.5 and 0.25 MN in the x-direction and all other entries of b are zero.

The 12×12 matrix A is exactly symmetric, and 86 of its 144 entries are nonzero. The zeros couple nodes that share no triangle, such as nodes 1 and 7. Every entry is an integer multiple of $Et/(40(1-\nu^2)) \approx 5.49 \times 10^7$ N/m; the full matrix is in the repository file `draft/experiments/results/square/small_A_integer.csv`. Its eigenvalues range from 1.07×10^8 to 1.03×10^{10} N/m, so $\kappa_2(A) = 95.8$ and the system is well conditioned.

Pivoted $PA = LU$, the explicit inverse and Cholesky all give relative residuals η (Eq. (12)) below 10^{-15}, at rounding level. The right-edge nodes move $u_x = 0.511$, 0.482 and 0.468 mm from bottom to top, already close to 0.5 mm on this coarse mesh.

4.2 A larger system and the iterative methods

For $m = 20$ the mesh has 441 nodes, 800 triangles and $N = 840$ unknowns. Now $\kappa_2(A) = 9104$, and the scaled matrix S of Eq. (10) has largest eigenvalue 2.96, so $\rho_J = 1.96$. Table 1 and Fig. 3 compare all methods on this system.

The simple iterations behave as Section 3 predicts. Since $\rho_J = 1.96 > 1$, one error component of Jacobi nearly doubles and changes sign each sweep, so the method diverges and is stopped

Table 1: All methods on the 840-unknown system (Section 3 protocol, best of three times). An asterisk marks a method that did not reach $\eta < 10^{-8}$ within 20,000 iterations.

Method	Iterations	Final η	Error vs LU	Time (ms)
Sparse LU	–	3.5×10^{-14}	–	2.3
Jacobi*	49	1.3×10^{10}	3.4×10^{7}	0.65
Damped Jacobi ($\omega = 0.5$)*	20,000	4.8×10^{-6}	8.7×10^{-5}	284
Gauss–Seidel	19,136	9.996×10^{-9}	1.7×10^{-7}	7,180
Steepest descent*	20,000	3.1×10^{-6}	4.2×10^{-5}	541
CG	242	8.8×10^{-9}	9.6×10^{-10}	5.4
Diagonal PCG	226	8.8×10^{-9}	4.9×10^{-10}	6.8
MINRES	236	9.4×10^{-9}	5.9×10^{-9}	13.7
Full GMRES	228	9.5×10^{-9}	6.0×10^{-9}	186

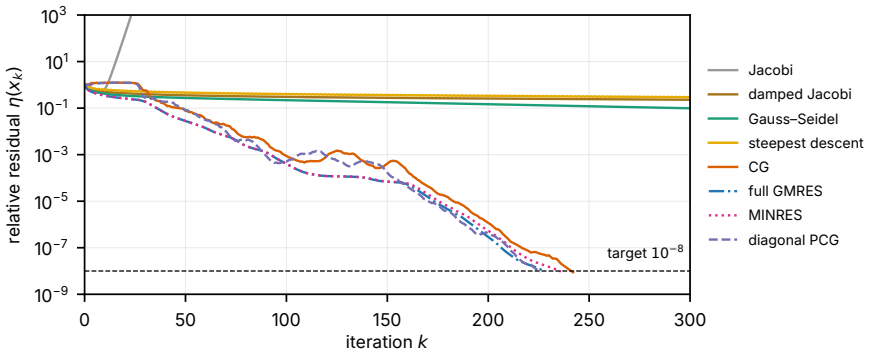


Figure 3: Relative residual $\eta(x_k)$ over the first 300 iterations on the 840-unknown system, with the target 10^{-8} dashed. The Krylov methods reach it in 226–242 iterations; the other methods have barely started.

after 49 sweeps. Damped Jacobi and steepest descent converge, but too slowly to reach the target within 20,000 iterations. Gauss–Seidel reaches it only after 19,136 sweeps and 7.2 s.

CG needs 242 iterations. Bound (11) with $\kappa = 9104$ guarantees the target within about $\frac{1}{2}\sqrt{\kappa}\,\ln(2 \times 10^{8}) \approx 912$ iterations, so CG does about four times better than its worst case [11]. Diagonal preconditioning saves only 16 iterations, because the diagonal of A is nearly constant on this uniform mesh.

Full GMRES uses about as many iterations as CG but takes 35 times longer. It ignores the symmetry of A and must store and orthogonalize against the whole Krylov basis, so each step costs more than the last [7]. At this small size, sparse LU (2.3 ms) is faster than every iterative method.

4.3 Deformation and validation

Figure 2c shows the 20×20 solution. The right edge moves between 0.492 and 0.497 mm, close to the expected 0.5 mm. Near the loaded end the top and bottom edges move inwards by about 0.08 mm each, matching the free lateral contraction $\nu pL/E = 0.15$ mm of the whole width. The clamp prevents this contraction at $x = 0$, which gives the tapered shape.

The solution passes four checks. All 42 prescribed components are exactly zero. The reactions of Eq. (7) balance the applied force of 10^{6} N and moment of 5×10^{5} N m to within 5×10^{-9} N and N m. The five methods that met the target agree with LU to a relative error of at most 1.7×10^{-7}. Finally, the patch test of Section 3 on the same mesh reproduces the exact linear field to a relative error of 2.6×10^{-14}, which confirms assembly, loading and reduction. The clamped plate has no exact solution; Section 6 estimates its discretization error.

5 Additional objects and experimental design

The square of Section 4 stretches almost uniformly and has a known answer. Real parts have holes, corners and slender members. This raises three questions: how shape changes the deformation, whether it changes A and hence the cost of solving, and whether results from a regular grid carry over to an irregular mesh. We set up four objects and a list of comparisons, each changing one factor at a time; Section 6 reports the outcomes.

Figure 4 shows the objects, with lengths in metres. The square S is the plate of Section 4 on an irregular mesh. The plate H adds a central hole of radius 0.15; in a wide plate under tension, the stress at a small hole is three times the applied stress [13]. The bracket L has counterclockwise vertices $(0,0)$, $(1,0)$, $(1,0.5)$, $(0.5,0.5)$, $(0.5,1)$, $(0,1)$. At its inner corner the exact stress is infinite [16], so stress there cannot settle under refinement. The beam B, $0 \le x \le 4$, $0 \le y \le 0.5$, bends, which linear triangles represent poorly on coarse meshes [2].

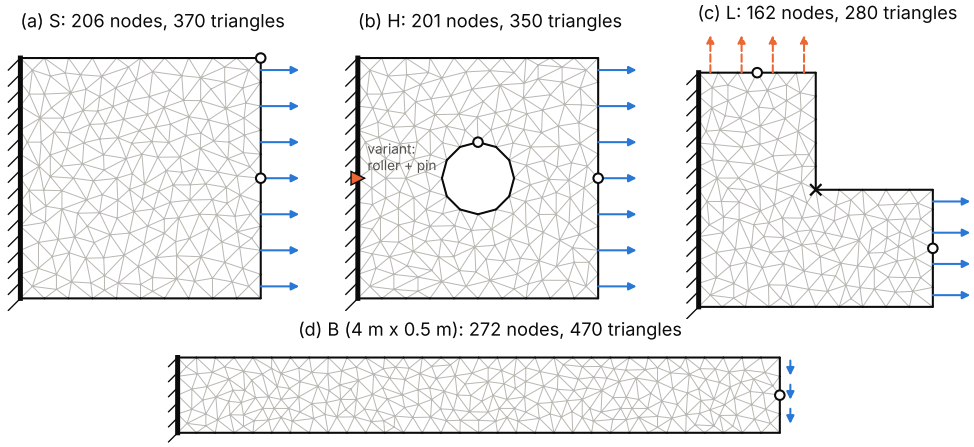


Figure 4: The four objects on their coarsest meshes. Hatching: clamped edge. Blue arrows: baseline load. Dashed orange arrows: alternative load on L. Orange triangle: pin used with a roller support on H. Circles: points where displacements are compared; cross: inner corner of L.

We mesh each object with Triangle [10]. Its triangles follow the boundary, leave the hole empty and have no angle below 30°. Four levels, with mesh size h from 0.1 to 0.025 (beam: 0.125 to 0.031), give $N \approx 300$ to 7800 unknowns. The comparison points are nodes on every level, and we check that triangle areas are positive and add up to the area of the region.

Material and thickness are those of Section 4. Every object is clamped on $x = 0$, which stops both translations and the rotation, so A is symmetric positive definite; we confirm that its smallest eigenvalue is positive. S, H and L carry the traction $p = 100$ MPa on their face $x = 1$. The beam carries a downward tip traction of 10 MPa, a force $P = 50$ kN; beam theory predicts a tip deflection $PL_B^3/(3EI) = 51.2$ mm, with length $L_B = 4$, $I = tW_B^3/12$ and depth $W_B = 0.5$ [13]. Loads act on the same boundary pieces on every mesh, so each level carries the same total force.

Table 2 lists the comparisons. Several have known answers and so test the code: scaling the load scales the displacement but not the iteration counts; scaling E multiplies A by a constant, leaving $\kappa_2(A)$ unchanged; moving the load on L, or turning it on B, changes b but not A. Changing ν or the supports changes A itself. Under refinement, changes at the comparison points estimate the discretization error, while κ_2 and iteration counts show how solver cost grows with N.

Every system is solved by sparse LU, CG, diagonal PCG, MINRES and Gauss–Seidel under the rules of Section 3, and the second level of each object by all methods. As in Section 4, loads and support reactions must balance in force and moment.

Table 2: Case register. Variants of H, L and B use the second mesh level. u_P: displacement at a comparison point; iter.: iteration counts.

Study	Question	Changed	Fixed	Measured
Geometry	effect of shape	S, H, L	load, clamp, h	u_P, κ_2, iter.
Refinement	convergence, cost	h, 4 levels	object, load	u_P, stress, κ_2, iter.
Mesh type	grid vs irregular	S vs grid, $m = 10$–40	square problem	u_P, κ_2, iter.
Load size	linearity	$p = 25$, 100, 250 MPa (H)	A	u_P/p, iter.
Load location	position	face $x = 1$ vs $y = 1$ (L)	A, total force	u_P, iter.
Supports	clamp vs roller	roller on $x = 0$ + pin (H)	load, mesh	u_P, κ_2, iter.
Material	role of E, ν	$\nu = 0$, 0.3, 0.45; $E = 70$ GPa (H)	shape, load	u_P, κ_2, iter.
Bending	beam accuracy	h, 4 levels (B)	beam, P	tip u_y vs theory, κ_2
Load type	axial vs bending	tip load direction (B)	A	tip u, iter.

6 Results, validation and discussion

We now report the comparisons of Table 2; displacements refer to the points marked in Fig. 4.

Physical response. Under the square's load, the loaded face of the plate with a hole (Fig. 5) moves 0.676 mm instead of 0.492 mm: removing material makes the plate softer. The load must flow around the hole, so stress gathers at its top and bottom. The largest element stress σ_{xx} there rises from $2.80p$ to $3.68p$ over the four meshes and has not yet settled. At the inner corner of the bracket the displacement settles (0.124, 0.127, 0.128, 0.128 mm), but the largest stress in the adjacent triangles keeps growing (119, 138, 154, 178 MPa). This is expected, since the exact stress at an inward-pointing corner is infinite [16].

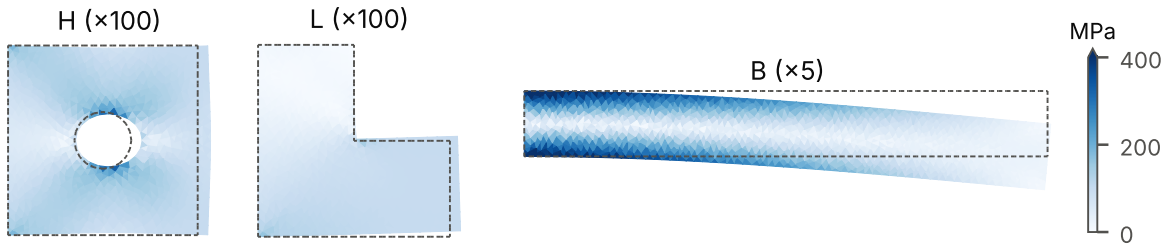


Figure 5: Deformed objects (second mesh level) coloured by element von Mises stress; dashed: original outline. Displacements are amplified 100 times (H, L) and 5 times (B).

The same 0.5 MN applied to the upper arm of the bracket moves the loaded face 0.304 instead of 0.472 mm and halves the corner stress (71 instead of 138 MPa), but the peak stress moves to the clamp and reaches 332 MPa. The beam is far more flexible in bending than in stretching: its tip moves 50.8 mm under a 50 kN transverse load but only 2.00 mm under a 500 kN axial load. A roller and pin instead of the clamp on H raise the displacement by only 0.9% but $\kappa_2(A)$ by 38%: fewer held components make the softest mode softer.

Displacements scale exactly with the load ($p = 25$ to 250 MPa) and with $1/E$ ($E = 70$ GPa), while stresses and $\kappa_2(A)$ are unchanged, since E only scales A. Poisson's ratio changes the shape of D: raising ν from 0 to 0.45 lowers the displacement by 1.7% but raises $\kappa_2(A)$ by 37% and the CG iterations from 278 to 332.

Mesh refinement and discretization error. Under refinement the matched-point displacements settle. For the square the change between levels falls from 4.3×10^{-4} to 5.0×10^{-5} mm, and at the loaded-face midpoint the finest irregular and structured meshes agree to 2×10^{-4} (relative). Near the hole, where stress varies rapidly, convergence is slower (changes from 2.3×10^{-2}

to 2.2×10^{-3} mm). The beam's tip deflection, 48.3, 50.8, 51.3 and 51.5 mm, approaches the beam-theory value of 51.2 mm, or 51.8 mm with shear deformation [13], from below: linear triangles are too stiff in bending [2]. The finest mesh is 0.7% below the estimate with shear. These changes, at least 10^{-4} relative, measure the discretization error. The algebraic error of every accepted CG solution against sparse LU is at most 1.0×10^{-9}, so the mesh, not the solver tolerance, limits the accuracy.

Conditioning and solver performance. Figure 6a shows that $\kappa_2(A)$ grows in proportion to N for every object, with fitted log–log slopes 0.98–1.02. This is the growth like h^{-2} expected for finite element stiffness matrices, since $N \propto h^{-2}$ in two dimensions [3]. By Eq. (11), CG iterations should then grow like $\sqrt{\kappa_2} \propto N^{1/2}$; the measured slopes are 0.50–0.52 (Fig. 6b). Shape changes the constant, not the rate. The beam's κ_2 is about 260 times the square's at similar N: bending a slender beam takes little energy, so its smallest eigenvalue is 290 times smaller. Yet CG needs only 1.8 times as many iterations, not $\sqrt{260} \approx 16$ times: only four beam eigenvalues lie below the square's smallest, and CG removes a few isolated eigenvalues in a few extra steps [11].

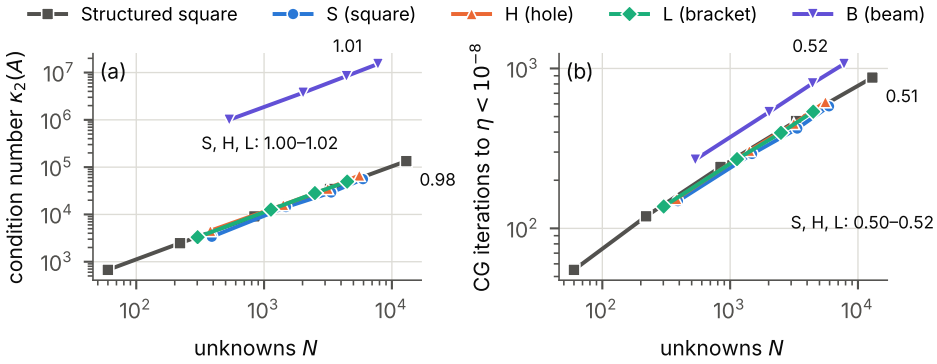


Figure 6: (a) $\kappa_2(A)$ and (b) CG iterations against N, with fitted log–log slopes; the structured square is that of Section 4.

Table 3: Solvers on the second mesh level: iterations / best time in ms. *: $\eta < 10^{-8}$ not met.

	S, $N = 1484$	H, $N = 1414$	L, $N = 1132$	B, $N = 2020$
$\kappa_2(A)$	1.43×10^4	1.60×10^4	1.27×10^4	3.75×10^6
Sparse LU	4.6	4.6	2.9	5.0
CG	292 / 10.8	306 / 11.1	271 / 6.9	534 / 24.0
Diagonal PCG	264 / 10.6	281 / 12.2	246 / 9.5	493 / 20.9
MINRES	285 / 26.0	302 / 26.7	268 / 20.5	541* / 50.5
Gauss–Seidel	20,000* / 11,700	20,000* / 11,700	20,000* / 10,600	20,000* / 14,800

In Table 3, diagonal PCG saves only 4–12% of the CG iterations and no time: all triangles have similar size and the material is uniform, so the diagonal of A varies little and scaling by it lowers κ_2 by only 12–31%. Jacobi diverges on every mesh, as ρ_J of Eq. (10) is 1.09–1.44. Gauss–Seidel reaches the target only on the coarsest meshes of S, H and L; on the beam it is still at $\eta \approx 0.2$ after 20,000 sweeps. Sparse LU is the fastest method on every system, including the largest ($N = 7776$: 31 ms against 162 ms for CG).

MINRES fails the test on all four beam meshes (final η from 1.6×10^{-8} to 1.5×10^{-6}), yet differs from LU by at most 6×10^{-11}. In floating point, the residual of a very ill-conditioned system can stall at a level that grows with κ_2 times the machine precision, and the internal residual estimate of MINRES can drift from the true one [12]. MINRES thus stopped above the target: a failure under our protocol, though the answer is accurate.

Validation and sensitivity. Prescribed displacements are exactly zero in every case. Applied loads and the reactions of Eq. (7) balance in force and moment to a relative error below 5×10^{-14} for the plates and 3×10^{-10} for the worse-conditioned beam. Every accepted iterative solution agrees with LU to 1.4×10^{-7} or better. Held only horizontally, H can slide vertically: then $\lambda_{\min}(A)/\lambda_{\max}(A) \approx 10^{-16}$ and Cholesky fails, whereas every supported case has $\lambda_{\min} > 0$. A random load perturbation of relative size 10^{-6} on the 20×20 structured square ($\kappa_2 = 9104$) changed the displacement by 4.5×10^{-7}, far inside the bound 9.1×10^{-3} of Eq. (13).

Limitations. We tested only diagonal preconditioning, not incomplete Cholesky or multigrid. Element stresses are constant per triangle, so peaks depend on the mesh. The beam (up to 540 MPa) and the 250 MPa case exceed the yield stress of steel, which the linear model ignores. Timings come from one machine and implementation; only ratios and trends are meaningful.

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